Limits & Fits

Limits

A single dimension controlling the size of a part can be specified in two ways.

Unilateral limits are written like $25^{+0.18}_{+0.10}$, giving the basic size 25, the upper limit 25.18, the lower limit 25.10, and a tolerance of 0.18-0.1 = 0.08mm

Bilateral limits are written as 25 ± 0.1, equivalent to $25^{+0.1}_{-0.1}$

Fits

When we mate two parts together, the relationship of the difference in sizes is called the fit.

Consider a shaft that has to fit in a hole.

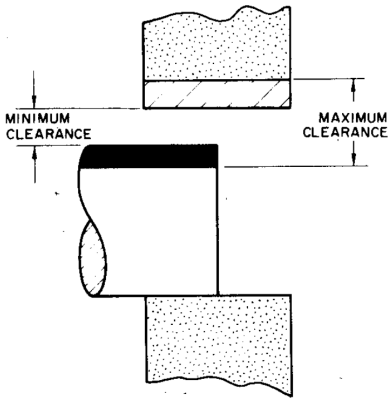
Firstly, the maximum possible gap can be calculated by the difference of the largest possible hole diameter by the smallest possible shaft diameter. If the difference is positive, we call this the maximum clearance, If negative, the parts overlap, and it is called the maximum interference.

Similarly, the smallest possible gap is the difference of the smallest possible hole diameter by the largest possible shaft diameter. The sign of the difference similarly explains if it is a clearance of inteference.

There are three types of fits of parts.

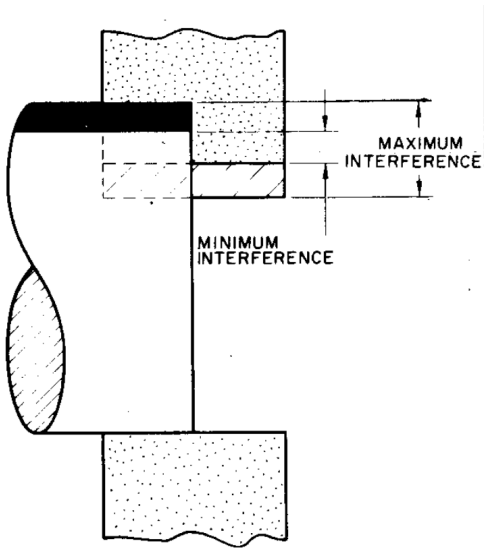
Clearance Fits

Two parts have a clearance fit if the minimum and maximum gap are both clearances:



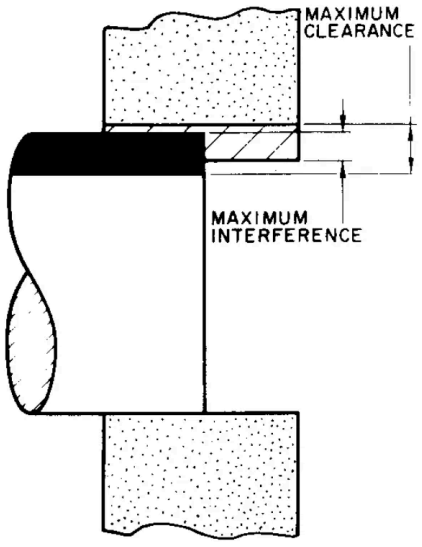
Interference Fits

Two parts have an interference fit if both the minimum and maximum gaps are interferences:



Transition Fits

Two part have a transition fit if the maximum gap is a clearance but the minimum gap is an interference.



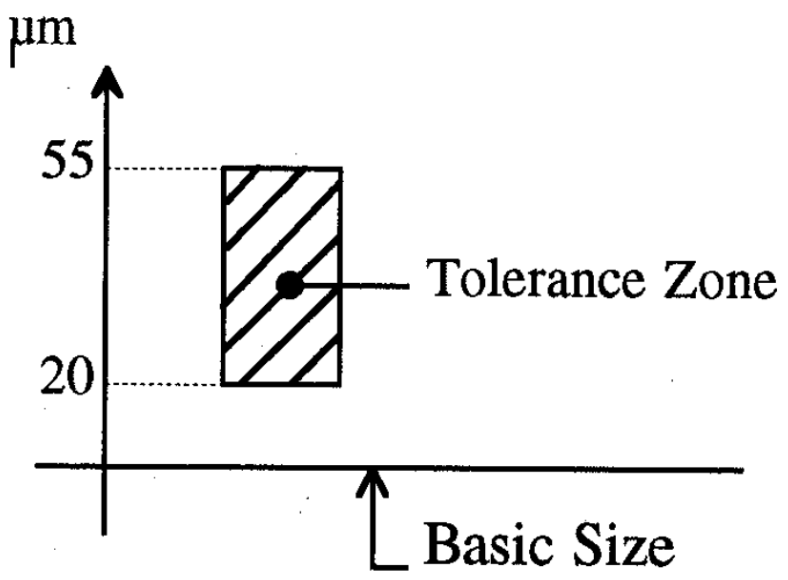
Terminology

To recap, the following terms are used for limits of a single part (say, $20^{+0.03}_{-0.01}$):

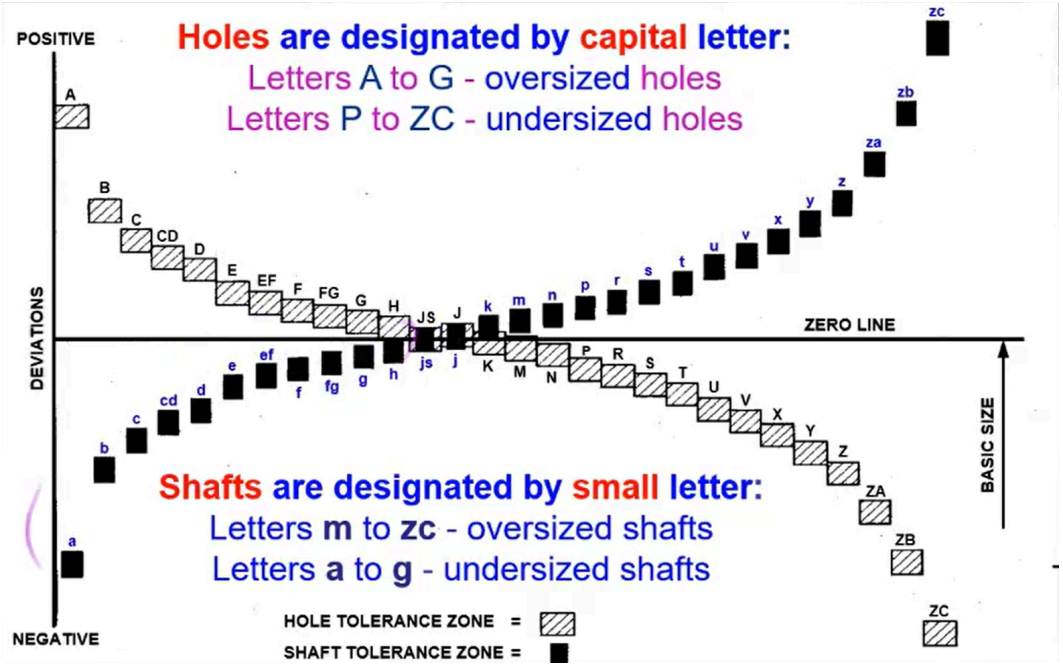
Name	Definition	Example
Basic Size	Nominal dimension	20
Max limit	maximum possible size	20.03
Min Limit	smallest possible size	19.99
Upper deviation	how much the upper limit deviates	0.03
Lower deviation	how much the lower limit deviates	-0.01
Tolerance	the total range of the deviation of the size	0.04

Tolerance Zones

Sometimes, the tolerance may be represented visually with a tolerance zone, representing the deviation from the basic size:



This is most commonly used for fundamental tolerances, which can be seen from the diagram:



where holes are represented with capital letters, and shafts are represented by lowercase letters. Notice how letters A to G have the limits above the basic size, hence the hole is oversized. Likewise, letters P to ZC have the limits below the basic size, hence the hole is undersized.

A similar convention applies for shafts too.

Tolerance Grades

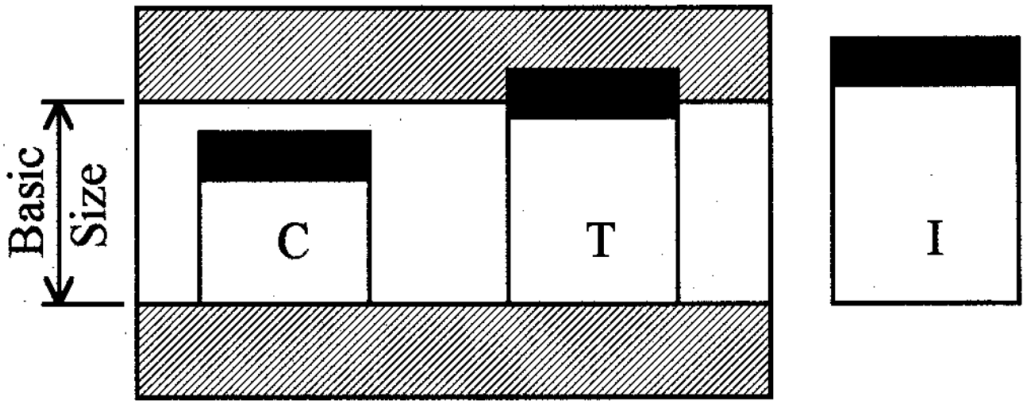
Tolerance grades represent the magnitude of the tolerance (the difference between deviations), in the system of IT01 to IT16. A lower number represents a smaller tolerance, which becomes more expensive to control.

Basis of Fits

Basis of fits are different methods used to control the type of fit between two parts during manufacturing.

Hole Basis Fits

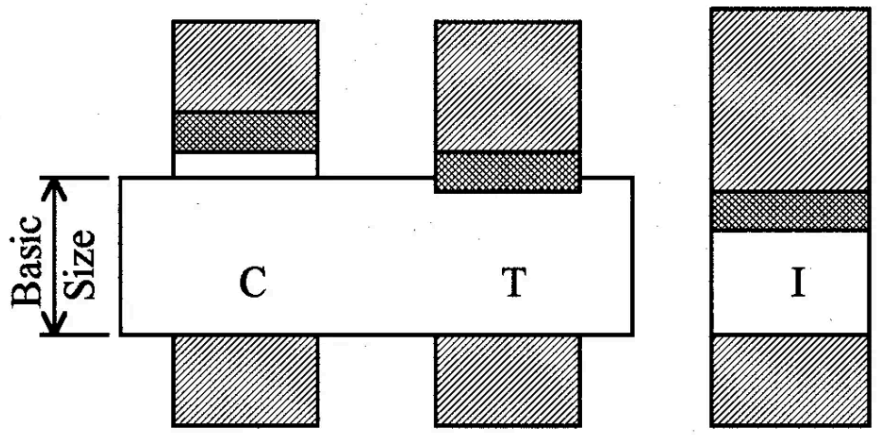
The fit of two parts is controlled by the shaft, with the basic hole size staying constant while



This allows for the fit to be controlled by changing the size of the shaft, allowing for more economic production as only one drill/reamer size is needed, and the shaft size can be adjusted via turning and grinding. Hence, this method is usually recommended, except when temperature causes size effects on larger parts

Shaft Basis Fits

The fit of two parts is controlled by the hole size, with the shaft size being constant:



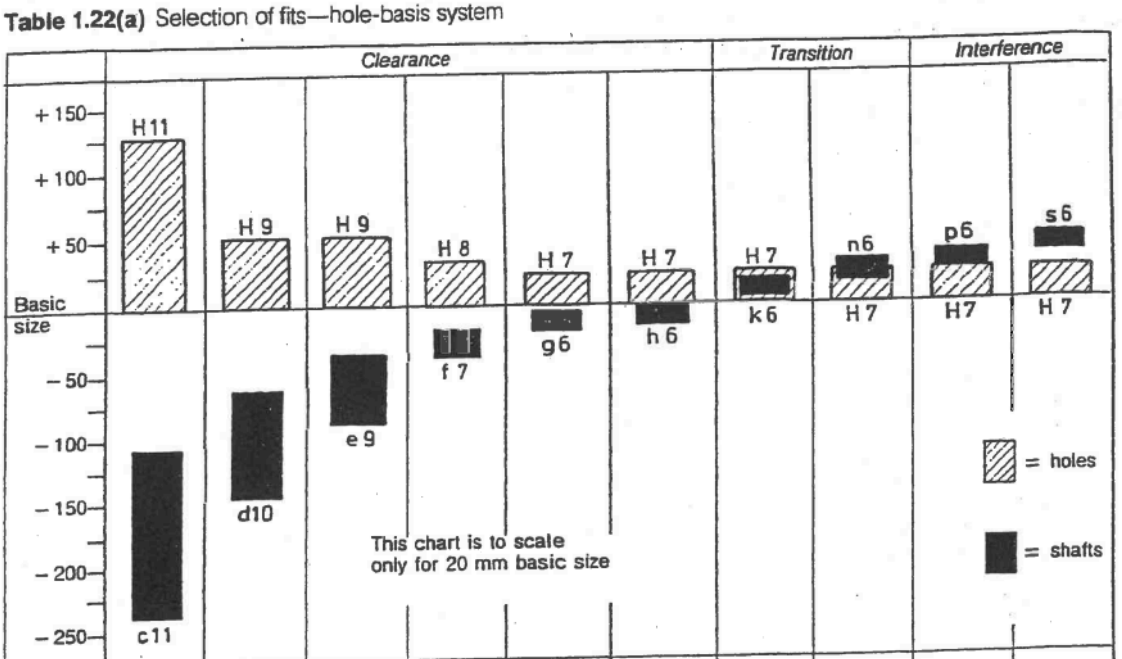
Legends: C - Clearance, T -Transition, I - Interference

Since multiple different drills and reamers are needed to control this fit, this is less common, however is still useful when multiple fits are needed across a long shaft, where it is easiest to keep a constant shaft diameter and vary the bores of the fitted parts.

ISO Fits

ISO fits are standard tolerances used to control a shaft/hole tolerance depending on the basic size. Shown is an ISO shaft/hole fit chart for the different tolerances, represented by a letter number pair.

Table 1.22(a) Selection of fits—hole-basis system



Tolerance unit = 0.001mm		Coarse tolerance	Loose running fit	Easy running fit	Normal running	Precision running, location	Average location	Light push fit	Heavy push fit	Press fit (ferrous)	Heavy press fit (non-ferrous)
Basic sizes (mm) over	to	H11–c11	H9–d10	H9–e9	H8–f7	H7–g6	H7–h6	H7–k6	H7–n6	H7–p6	H7–s6
0	3	+60 −60 0 −120	+25 −20 0 −60	+25 −14 0 −39	+14 −6 0 −16	+10 −2 0 −8	+10 0 0 −6	+10 +6 0 0	+10 +10 0 +4	+10 +12 0 +6	+10 +20 0 +14
3	6	+75 −70 0 −145	+30 −30 0 −78	+30 −20 0 −50	+18 −10 0 −28	+12 −4 0 −12	+12 0 0 −8	+12 +9 0 +1	+12 +16 0 +8	+12 +20 0 +12	+12 +27 0 +19
6	10	+90 −80 0 −170	+36 −40 0 −98	+36 −25 0 −61	+22 −13 0 −28	+15 −5 0 −14	+15 0 0 −9	+15 +10 0 +1	+15 +19 0 +10	+15 +24 0 +15	+15 +32 0 +23
10	18	+110 −95 0 −205	+43 −50 0 −120	+43 −32 0 −75	+27 −16 0 −34	+18 −6 0 −17	+18 0 0 −11	+18 +12 0 +1	+18 +23 0 +12	+18 +29 0 +18	+18 +39 0 +28
18	30	+130 −110 0 −240	+52 −65 0 −149	+52 −40 0 −92	+33 −20 0 −41	+21 −7 0 −20	+21 0 0 −13	+21 +15 0 +2	+21 +28 0 +15	+21 +35 0 +22	+21 +48 0 +35
30	40	+160 −120 0 −280	+62 −80 0 −180	+62 −50 0 −112	+39 −25 0 −50	+25 −9 0 −25	+25 0 0 −16	+25 +18 0 +2	+25 +33 0 +17	+25 +42 0 +26	+25 +59 0 +43
40	50	0 −130 −290									
50	65	+190 −140 0 −330	+74 −100 0 −220	+74 −60 0 −134	+46 −30 0 −60	+30 −10 0 −29	+30 0 0 −19	+30 +21 0 +2	+30 +39 0 +20	+30 +51 0 +32	+30 +72 0 +53
65	80	0 −150 −340									+78 +59
80	100	+220 −170 0 −390	+87 −120 0 −260	+87 −72 0 −159	+54 −36 0 −71	+35 −12 0 −34	+35 0 0 −22	+35 +25 0 +3	+35 +45 0 +23	+35 +59 0 +37	+35 +93 0 +71
100	120	0 −180 −400									+101 +79
120	140	+250 −200 0 −450	+100 −145 0 −305	+100 −84 0 −185	+63 −43 0 −83	+40 −14 0 −39	+40 0 0 −25	+40 +28 0 +3	+40 +52 0 +27	+40 +68 0 +43	+40 +117 0 +92
140	160	0 −210 −460									+125 +100
160	180	0 −230 −480									+133 +108
180	200	+290 −240 0 −530	+115 −170 0 −355	+115 −100 0 −215	+72 −50 0 −96	+46 −15 0 −44	+46 0 0 −29	+46 +33 0 +4	+46 +60 0 +31	+46 +79 0 +50	+46 +151 0 +122
200	225	0 −260 −550									+159 +130
225	250	0 −280 −570									+169 +140
250	280	+320 −300 0 −620	+130 −190 0 −400	+130 −110 0 −240	+81 −56 0 −108	+52 −17 0 −49	+52 0 0 −32	+52 +36 0 +4	+52 +66 0 +34	+52 +88 0 +56	+52 +190 0 +158
280	315	0 −330 −650									+202 +170
315	355	+360 −360 0 −720	+140 −210 0 −440	+140 −125 0 −265	+89 −62 0 −119	+57 −18 0 −54	+57 0 0 −36	+57 +40 0 +4	+57 +73 0 +37	+57 +98 0 +62	+57 +226 0 +190
355	400	0 −400 −760									+244 +208
400	450	+400 −440 0 −840	+155 −230 0 −480	+155 −135 0 −290	+97 −68 0 −131	+63 −20 0 −60	+63 0 0 −40	+63 +45 0 +5	+63 +80 0 +40	+63 +108 0 +68	+63 +272 0 +232
450	500	0 −480 −880									+292 +252

General Tolerances

Tolerances should only be applied to dimensions if it is critical, otherwise more work has to be done to control the dimension. Instead, general tolerances are used for non-critical dimensions.

General tolerances may either be constant for any other dimension, or change depending on the dimension size. These tolerances are usually selected from ISO standards, like the ones shown:

Table 1. Permissible variations for linear dimensions

Nominal dimensions, mm		0.5 up to 3	Over 3 up to 6	Over 6 up to 30	Over 30 up to 120	Over 120 up to 315	Over 315 up to 1000	Over 1000 up to 2000
		mm	mm	mm	mm	mm	mm	mm
Permissible variations	Fine series	± 0.05	± 0.05	± 0.1	± 0.15	± 0.2	± 0.3	± 0.5
	Medium series	± 0.1	± 0.1	± 0.2	± 0.3	± 0.5	± 0.8	± 1.2
	Coarse series	–	± 0.2	± 0.5	± 0.8	± 1.2	± 2	± 3

Table 2. Permissible variations for angular dimensions

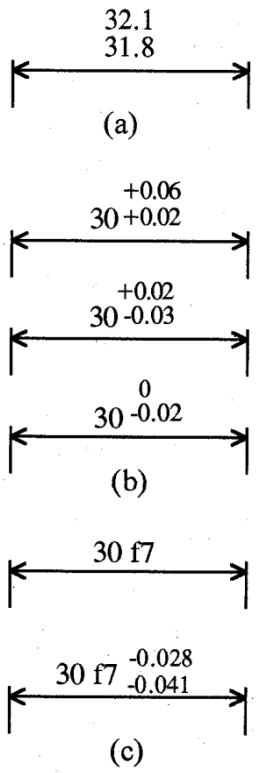
Length of the shorter side, mm		Up to 10	Over 10 up to 50	Over 50 up to 120	Over 120 up to 400
Permissible variations	in degrees and minutes	± 1°	± 30′	± 20′	± 10′
	in millimetres per 100 mm	± 1.8	± 0.9	± 0.6	± 0.3

Tolerance Specifications on Drawings

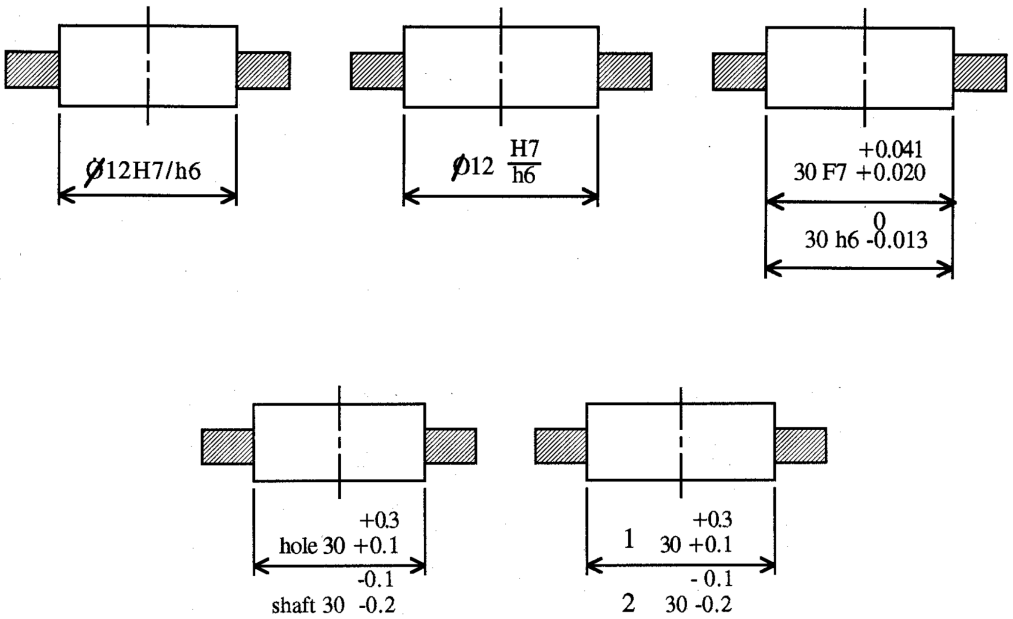
On drawings, there are different ways to specify the same tolerance.

For single linear dimensions, they can be represented by

- The maximum and minimum limits, reducing calculations during machining
- The deviations
- The ISO tolerance, optionally with the deviations

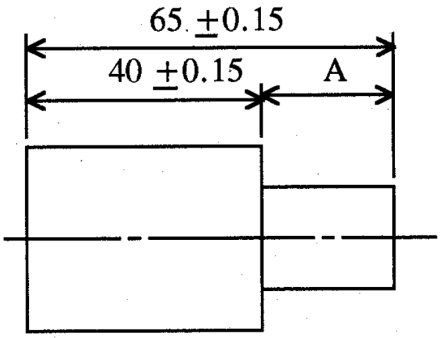


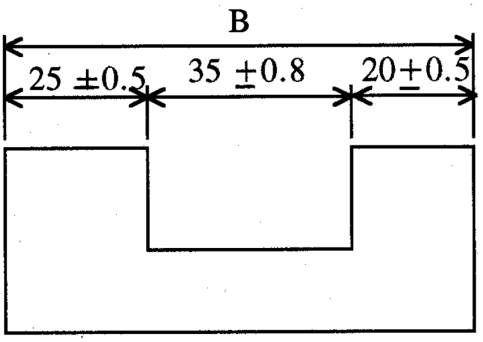
For assemblies, the basic size is written followed by the grade, optionally with the deviations, or is just written with only the deviations:



Tolerance Accumulation Analysis

Depending on the way a part is dimensioned, tolerances can stack and accumulate, which may cause a higher tolerance in other dimensions, which may interfere with other parts during mating.





$A = 65 \pm 0.15 - 40 \pm 0.15$
$= 25 \pm 0.3$
A max. = 25.3 mm
A min = 24.7 mm

$B = 25 \pm 0.5 + 35 \pm 0.8 + 20 \pm 0.5$
$= 80 \pm 1.8$
B max. = 81.8 mm
B min = 78.2 mm

Generally, the tolerance of the undimensioned length is found by the sum of the other tolerances, which shows that these dimensions are harder to control. Hence, crucial dimensions should be toleranced directly to ensure they work for their intended purposes.